Probability and Statistics
Topic 4. Discrete Random Variables

- Random variable
- Discreate random variable
- Probability mass function
- Cumulative distribution function
- Bernoulli distribution
- Binomial distribution
- Geometric distribution
- Poisson distribution
- Joint probability mass function
- Marginal probability mass function
- Conditional distribution
- Independent random variables

Seminar 4

Problem 1 (Discrete r.v. and their p.m.f.)**.** Suppose that a fair die is rolled twice. What is the probability mass function of the following random variables:

(a) $X =$ ‘‘`the maximum value to appear in the two rolls`’’;

(b) $Y =$ ‘‘`the minimum value to appear in the two rolls`’’. Give a short explanation why $p_X(j) = p_Y(7-j)$ for $j = 1, 2, \ldots, 6$.

(c) Find the joint p.m.f. of the random vector (X, Y); calculate the marginal distribution of X and compare to part (a).

(d) Are random variables X and Y independent?

Problem 2 (Joint probability distribution)**.** A fair coin is tossed three times. Denote by X the number of Heads in the first two tosses and by Y the number of Tails in the last two tosses of the coin.

(a) Find the joint probability mass function $p_{X,Y}$ of X and Y:

Events	$X=0$	$X=1$	$X=2$
$Y=0$	...	...	...
$Y=1$	...	...	...
$Y=2$	...	...	...

prob.	$X=0$	$X=1$	$X=2$
$Y=0$	...	...	...
$Y=1$	...	...	...
$Y=2$	...	...	...

(b) Calculate the marginal p.m.f.'s of X and Y

(c) What distributions do X and Y have?

(d) Are X and Y independent?

(e) Find the probability that X is equal to Y); that X is greater than Y

(f) Calculate the conditional p.m.f. of X given that $Y = 1$

Problem 3 (Poisson distribution)**.** Suppose that the number of new COVID-19 positive tested patients in the city A on day i is a Poisson random variable X_i of parameter $\lambda = 10$; assume also that the random variables $X_1, X_2, \dots$ are independent.

(a) Find the probability that the number of newly COVID-19 infected patients tomorrow will be larger than today's count.

Assume that the number of new COVID-19 positive tested patients in the city B on day i is a Poisson random variable Y_i of parameter $\mu = 20$; assume also that all random variables X_i and Y_j, $i, j = 1, 2, \dots$ are independent.

(b) Show that the random variable $X_i + Y_i$ has Poisson distribution of parameter 30. In the general case, if X and Y are independent Poisson random variables with parameters λ and μ respectively, what is the distribution of the random variable $X + Y$?

(c) Assume that by this noon, 5 new cases were reported. Find the probability that k newly COVID-19 infected patients will be reported today in total.
Hint: Can you see that there are two ways to solve this problem, depending on whether we make some assumptions about the distribution of cases throughout the day? What assumptions can be made?

(d) Assume that 30 new COVID-19 infected patients were reported yesterday in the cities A and B in total. What is the probability that k of them come from A, $k = 0, 1, \dots, 30$?

Problem 4 (Binomial distribution)**.** A communications channel transmits the digits 0 and 1. However, due to noise, the digit transmitted is incorrectly received with probability 0.2. Suppose that we want to transmit an important message consisting of n binary digits. To reduce the chance of error, we transmit 00000 instead of 0 and 11111 instead of 1. If the receiver of the message uses the "majority" rule for decoding, what is the probability that

(a) the message will be wrong when decoded?

(b) at most two binary digits of the original message will be wrongly decoded?

(c) what is the most probable number of correctly decoded binary digits?

What independence assumptions are you making?

Problem 5 (Discrete r.v. and their p.m.f.)**.** Five men and five women are ranked according to their scores on an examination. Assume that no two scores are alike and all 10! possible rankings are equally likely. Let X denote the highest ranking achieved by a woman and let Y denote their lowest ranking. (For instance, $X = 1$ if the top-ranked person is a female.)

(a) Find the probability mass function (p.m.f.) of a random variable X;

(b) explain how to find the joint p.m.f. of a random vector (X, Y), i.e., how to calculate the values $p_{(X,Y)}(x, y) := \mathsf{P}(X = x, Y = y)$ for all pairs of x and y;

(c) what is the marginal distribution of X? Calculate the value of the marginal distribution of X for $x = 3$ and compare to the value found in part (a);

(d) are the random variables X and Y independent?

Problem 6 (Binomial distribution)**.** When coin A is flipped, it lands on Heads with probability 0.4; when coin B is flipped, it lands on Heads with probability 0.6. One of these coins is randomly chosen and flipped 10 times; let X denote the total number of Heads.

(a) Is it true that a r.v. X has a binomial distribution?

(b) Find the p.m.f. of the random variable X.

(c) Given that the first of these ten flips lands Heads, what is the conditional distribution of the total number of Heads? (Hint: condition on the coin chosen)

(d) Assuming that $X = k$, what is the (conditional) probability that coin A is tossed? For what k is it larger than 0.5?

Problem 7 (Binomial distribution)**.** Suppose that a biased coin that lands on Heads with probability p is flipped 10 times. Given that a total of 6 Heads results, find the conditional probability that the first 3 outcomes

(a) are HTT (meaning that the first flip results in Heads, the second in Tails, and the third in Tails);

(b) include precisely k Heads; $k = 0, 1, 2, 3$.

Assume now that there are 2 Heads in the first three tosses of the coin.

(c) What is the conditional probability that there are k Heads in all 10 tosses?

(d) What is the most probable number of Heads in all 10 tosses?

Problem 8 (Joint probability distribution)**.** You roll two fair dice, and they show X and Y, respectively. Let $U = \min\{X, Y\}$, $V = \max\{X, Y\}$.

(a) Write down the joint distributions of:

(i) $\{U, X\}$; (ii) $\{U, V\}$; (iii) $\{X, Y, V\}$

(b) Which of the above **r.v.** are independent?

(c) Find the conditional p.m.f. of the **r.v.** $X \mid U = u$ and $Y \mid V = v$.

Problems to be discussed during consultation

Problem 9. Two balls are chosen randomly from an urn containing 8 white, 4 black, and 2 orange balls. Suppose that we win 2 for each black ball selected and we lose 1 for each white ball selected. Let X denote our winnings. What are the possible values of X, and what are the probabilities associated with each value?

Problem 10. Let the random variables X and Y have joint probability mass function $f(x, y)$ such that:

$$f(1,2) = \tfrac{1}{8}, \qquad f(1,3) = \tfrac{1}{16}, \qquad f(1,4) = \tfrac{1}{4},$$
$$f(2,2) = \tfrac{1}{16}, \qquad f(2,3) = \tfrac{1}{8}, \qquad f(2,4) = \tfrac{3}{8}.$$

(a) Find the marginal **p.m.f.** of the **r.v.** X and Y;

(b) Find the probability of the following events:
(1) $X > Y$; (2) $X + Y$ is odd;
(3) $X \geq Y$; (4) $X - Y \leq 1$.

Problem 11. Suppose that the number of accidents occurring on a highway each day is a Poisson random variable with parameter $\lambda = 3$.

(a) Find the probability that 3 or more accidents occur today.

(b) Repeat part (a) given the information that 1 accident has already occurred today.

Problem 12. Five distinct numbers are randomly distributed to players numbered 1 through 5. Whenever two players compare their numbers, the one with the higher one is declared the winner. Initially, players 1 and 2 compare their numbers; the winner then compares her number with that of player 3, and so on. Let X denote the number of times player 1 is a winner. Find $\mathbb{P}\{X = i\}, i = 0, 1, 2, 3, 4$.